TEXORA · COURSE NOTES · LECTURE SHEET

Introduction to Textile Fibres

What a textile fibre is, the properties a material must have to earn the name, and the complete classification of fibres — natural and man-made.

TE-1209 · NATURAL TEXTILE FIBRES · TOPIC 1 · LEVEL 1 TERM 2

1. What is a textile fibre?

Every textile begins as fibres, and every property of the finished product — its strength, softness, absorbency, warmth, even how it must be washed — is largely decided by the fibres it contains. Fibre science is therefore the foundation on which spinning, weaving, dyeing and finishing all stand.

> **DEFINITION**
>
> A textile fibre is a unit of matter characterised by flexibility, fineness and a high ratio of length to thickness — the length being at least one hundred times the diameter — which is capable of being spun into yarn or made directly into fabric.

The definition hides three requirements, and a material that fails any one of them cannot serve as a textile fibre. It must be **long relative to its thickness**, otherwise it behaves as dust or granules and cannot be twisted into a strand. It must be **flexible**, so that it bends around its neighbours instead of snapping. And it must be **processable** — able to survive opening, carding, drafting and twisting, and then the tensions of weaving or knitting and of daily use.

Fibres reach the mill in two physical forms. **Staple fibres** are short, measurable lengths — millimetres to a few hundred millimetres — such as cotton, jute and wool, and man-made fibres cut to similar lengths. **Filaments** are continuous strands of indefinite length: silk as it comes from the cocoon, and all man-made fibres as they leave the spinneret, which may be used as filament yarn directly or cut into staple for spinning on fibre-spinning systems.

2. The essential properties of a textile fibre

Not every thin material is a fibre, and not every fibre is worth spinning. The properties divide into **primary (essential)** properties, without which a fibre cannot be spun at all, and **secondary (desirable)** properties, which decide comfort, appearance and care.

2.1 Primary properties

- **High length-to-diameter ratio.** At least about 100:1. This geometry lets fibres overlap along a yarn so that friction can develop between them.
- **Adequate strength (tenacity).** The fibre must withstand the tensions of spinning and weaving and the stresses of use. Strength is compared fairly between fibres of different fineness as *tenacity* — the breaking force divided by the linear density.
- **Flexibility.** The fibre must bend repeatedly without breaking, so it can be twisted, woven, knitted and worn.
- **Cohesiveness (spinnability).** Fibres must cling to one another when twisted, so the yarn does not fall apart. Surface roughness, crimp and natural convolutions (as in cotton) all help.
- **Uniformity.** Fibres of even length and fineness spin into even yarn with few breaks; wide variation makes weak, irregular yarn.

2.2 Secondary properties

- **Density** — decides how heavy a fabric of given thickness feels (polyester is denser than polypropylene; fabrics differ accordingly).
- **Moisture absorbency** — decides comfort against skin and how readily the fibre dyes; measured as *moisture regain*, the water a dry fibre absorbs from a standard atmosphere.
- **Elastic recovery** — the ability to return to shape after stretching, which governs creasing and fit.
- **Chemical and biological resistance** — behaviour toward alkalis, acids, sunlight, mildew and insects decides care requirements and service life.
- **Thermal behaviour** — whether the fibre melts, softens, chars or burns decides ironing safety and suitability for protective clothing.
- **Colour and lustre** — the natural appearance that processing must preserve or improve.

> **WORKED EXAMPLE — WHY COTTON QUALIFIES**
>
> Test cotton against the primary list. A cotton fibre is typically 20–50 mm long and only about 12–20 micrometres wide, so its length-to-diameter ratio is well over 1000:1. It has useful tenacity, higher when wet than dry — unusual and valuable for washing. It is flexible, and its ribbon-like shape with natural twists (*convolutions*) gives excellent fibre-to-fibre cohesion, so it spins easily even on simple machinery. Finally it is absorbent and comfortable. Cotton does not merely pass the test; it passes on almost every line, which is why it has been a principal textile fibre for thousands of years.

3. Classification of textile fibres

Fibres are classified first by origin — natural or man-made — and then by chemical family, because chemistry predicts behaviour (dyeing, washing, heat) better than any other single feature.

MAIN CLASS	SUB-CLASS	EXAMPLES AND NOTES
Natural	Vegetable (cellulosic)	**Seed:** cotton, kapok · **Bast:** flax (linen), jute, hemp, ramie · **Leaf:** sisal, abaca (manila) · **Fruit:** coir (coconut)
	Animal (protein)	**Hair:** wool (sheep), mohair, cashmere, camel hair · **Secretion:** silk (mulberry and wild/tussah)
	Mineral	Asbestos (historically important; now avoided for health reasons), natural mineral fibres of minor use
Man-made	Regenerated (from natural polymers)	Viscose rayon, lyocell, modal (from cellulose); acetate (modified cellulose); regenerated protein fibres of minor importance
	Synthetic (from petrochemical monomers)	Polyester, nylon (polyamide), acrylic, polypropylene, elastane (spandex) — polyester is the world's most-used fibre
	Inorganic	Glass fibre, carbon fibre, metallic fibres — mainly for technical textiles

Two points in this table deserve emphasis. First, the chemical division cuts across the natural/man-made one: viscose is man-made but chemically cellulose, so it dyes and absorbs almost like cotton; wool and silk are both proteins, so they share dye classes and a common weakness to alkali. When a later course explains dyeing, it will reason from this chemistry. Second, the **course TE-1209 is concerned with the natural fibres**: the cellulosic fibres in Topic 2, the protein fibres in Topic 3, and the remaining natural fibres in Topic 4.

2.3 A feel for the numbers: moisture regain

Moisture regain shows, in a single column of figures, why fibre choice is felt on the skin. The values below are the standard-condition regains (about 65% relative humidity, 20–21 °C) quoted in textile references, rounded for orientation:

FIBRE	MOISTURE REGAIN (APPROX.)	FIBRE	MOISTURE REGAIN (APPROX.)
Wool	13–16%	Cotton	7–8.5%
Silk	~11%	Flax (linen)	8–10%
Viscose	11–13%	Jute	~12.5%

Nylon	4–4.5%	Acrylic	1.5–2%
Polyester	~0.4%	Polypropylene	~0%

The order is no accident: protein and cellulosic fibres carry chemical groups that attract water, while polyester and polypropylene are hydrocarbons with almost nothing for water to hold on to. High regain means comfort and easy dyeing but slow drying and weight gain when wet; low regain means quick drying and dimensional stability but static build-up and a clammy feel in hot weather. Blend design — cotton with polyester, for instance — is largely the art of averaging this table.

4. Measuring fibres: the units you will meet everywhere

Fibre description becomes precise through a handful of measurements and their units — these recur in every later course, so fix them now.

- **Length** — quoted in millimetres (staple fibres) and classified as short, medium, long or extra-long staple; filaments are effectively infinite and are measured by weight instead.
- **Linear density (fineness)** — mass per unit length. The **tex** system: grams per 1,000 metres (a 1 tex fibre weighs 1 g per kilometre); its sub-unit **decitex (dtex)** is standard for fibres. The older **denier**: grams per 9,000 metres, still standard for silk and much filament yarn. Conversion: denier = 9 × tex. Finer fibre = smaller number in both systems.
- **Diameter** — wool and hair fibres are specified directly by mean diameter in micrometres (microns), because their near-round cross-section makes diameter meaningful; flat or irregular fibres (cotton, bast) are better described by linear density.
- **Tenacity** — breaking force per unit linear density, quoted in cN/tex (or grams per denier): the strength comparison that stays fair between a coarse jute strand and a fine silk filament.
- **Elongation at break** — the percentage stretch before rupture: wool stretches far (roughly 25–35% dry), cotton little (roughly 5–10%), and the figure predicts how forgiving a fabric feels in wear.

> **WORKED EXAMPLE — USING THE UNITS**
>
> A filament yarn is labelled 150 denier. In tex that is 150 ÷ 9 ≈ 16.7 tex. If its measured breaking force is 450 cN, its tenacity is 450 ÷ 16.7 ≈ 27 cN/tex. And a wool lot at 21 microns against one at 32 microns: the 21-micron lot is the apparel wool — finer, softer, and able to spin the finer yarn — before any other property is even discussed.

5. How fibres are identified

A first practical skill of the fibre technologist is to name an unknown fibre. The standard methods, in increasing order of certainty, are: **appearance and handle** (length, fineness, lustre, feel); the **burning test** (behaviour in flame, smell, and residue — cellulose burns like paper, protein smells of burning hair, most synthetics melt); **microscopy** (cotton's twisted ribbon, wool's scales and flax's cross-markings are unmistakable); **solubility tests** (each fibre family dissolves in a characteristic reagent); and instrumental methods such as infrared spectroscopy. Identification matters commercially — blends must be verified — and legally, because fibre content must be declared on garment labels.

6. Glossary — key terms of this topic

TERM	MEANING
Tenacity	The breaking strength of a fibre or yarn relative to its linear density; the fair way to compare strength between fine and coarse fibres.
Linear density	Mass per unit length of a fibre or yarn, expressed in systems such as tex or denier.
Moisture regain	The percentage of water absorbed by an oven-dry material from a standard atmosphere, calculated on the dry weight.
Staple fibre	A fibre of short, defined length, as distinct from a continuous filament.
Spinneret	The metal plate with fine holes through which a man-made fibre solution or melt is extruded to form filaments.
Bast fibre	A fibre obtained from the stem (phloem) of a plant, such as flax, jute, hemp or ramie.
Crimp	The waviness along a fibre's length, natural in wool and often added to man-made staple; it aids cohesion and bulk.

7. Key points and summary

- A textile fibre is defined by flexibility, fineness and a length-to-diameter ratio of at least about 100:1, together with the ability to be processed into yarn or fabric.
- The primary properties — ratio, tenacity, flexibility, cohesiveness and uniformity — decide spinnability; secondary properties such as absorbency, elasticity and chemical resistance decide comfort and care.
- Fibres are natural (cellulosic, protein, mineral) or man-made (regenerated, synthetic, inorganic). Chemistry, not origin alone, predicts dyeing and care behaviour.
- TE-1209 studies the natural fibres: cellulosic fibres next, then protein fibres, then the remaining natural fibres.

8. Practice

Answer in your own words; brief model answers follow.

1. A granule of plastic 0.5 mm long and 0.1 mm thick is offered as a “fibre”. Test it against the definition of a textile fibre and give your verdict.
2. Which primary property is missing in a fibre that is long and strong but perfectly smooth and straight, so that yarns made from it keep falling apart? Name a natural feature of cotton that supplies the property.
3. Convert: (a) a yarn of 30 tex into denier; (b) a silk thread of 45 denier into tex.
4. Two fibres break at the same force, but fibre B is twice as fine as fibre A. Which has the higher tenacity, and by what factor?
5. Classify each of these into the scheme of §3: viscose, nylon, jute, silk, glass fibre, kapok, acetate, mohair.
6. A summer shirt fabric must feel dry and cool; a winter base layer must feel warm even when damp. Using the regain table in §2.3, propose a fibre (or blend) for each and justify briefly.
7. An unknown white fibre burns like paper with almost no smell, and under the microscope shows a twisted ribbon. Identify it and name the two tests that decided you.
8. Explain why the mixed-partial rule “chemistry predicts behaviour” makes viscose dye more like cotton than like polyester, although viscose is man-made and polyester is too.

Answer key

1. Length-to-diameter ratio is 0.5 ÷ 0.1 = 5:1, far below the minimum of about 100:1 — it is a granule, not a fibre.
2. Cohesiveness (spinnability). Cotton’s natural convolutions give fibre-to-fibre grip; crimp in wool does the same.
3. (a) 30 × 9 = 270 denier. (b) 45 ÷ 9 = 5 tex.
4. Fibre B — tenacity is force per linear density, so at equal force and half the linear density, B’s tenacity is twice A’s.
5. Viscose — man-made regenerated (cellulosic); nylon — man-made synthetic; jute — natural cellulosic (bast); silk — natural protein; glass — man-made inorganic; kapok — natural cellulosic (seed); acetate — man-made regenerated (modified cellulose); mohair — natural protein (hair).
6. Shirt: cotton or linen (moderate regain, absorbent, cool handle). Base layer: wool (highest regain, absorbs moisture vapour while releasing a little heat, feels dry) — or a wool blend.
7. Cotton — burning behaviour of cellulose (like paper) plus the unmistakable convoluted ribbon under the microscope.

8. Dyeing follows chemistry, not origin: viscose is cellulose, like cotton, so it takes cotton dye classes and absorbency; polyester is a hydrophobic synthetic needing disperse dyes and heat.
